DAY 05 | POST-READ

APIs and FastAPI: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
HTTP request	Method + URL + headers + optional body.
HTTP response	Status code + headers + body (usually JSON).
REST	Resources as plural nouns, methods as verbs, stateless, consistent.
Methods	GET read, POST create, PUT replace, PATCH partial update, DELETE remove.
Safe and idempotent	GET is safe. GET, PUT, DELETE give the same result if repeated. POST does not.
Status codes	200 OK, 201 Created, 204 No Content, 400/422 bad data, 401/403 auth, 404 missing, 500 server error.
FastAPI	app = FastAPI(); @app.get("/path") decorators; return dicts as JSON.
Running	uvicorn main:app --reload, then open /docs.
Errors	raise HTTPException(status_code, detail).

Part B: Going beyond today

1. Idempotency explained

Pressing "pay" twice should not charge twice. PUT and DELETE are idempotent: repeating them leaves the same result. POST is not, so real payment APIs accept an idempotency key header to detect duplicates.

2. Query parameters

```
@app.get("/products")
def list_products(category: str | None = None, limit: int = 10):
    ...
# GET /products?category=home&limit=5
```

Path parameters identify one resource. Query parameters filter, sort or paginate a list. We go deeper on Day 6.

3. API versioning

Prefix routes with /v1 so you can change things later without breaking existing apps: /v1/products, then /v2/products.

4. Calling APIs from Python

```
import requests
r = requests.get("https://jsonplaceholder.typicode.com/todos/1")
print(r.status_code, r.json()["title"])
```

5. async def in FastAPI

FastAPI accepts both def and async def. Use async def when you call async libraries (such as an async database driver). Plain def is fine for everything else. We meet async properly on Day 10 with JavaScript.

6. CORS in one line

Browsers block a web page from calling an API on another address unless the API allows it. This is CORS. You will configure it when connecting React to FastAPI on Day 12.

Part C: Check yourself

- Which method would you use to change only the price of a product?
- What is the difference between 401 and 403?
- Why is /getProducts not a RESTful URL?
- What does --reload do when running uvicorn?
- Where does FastAPI's interactive documentation live?

Part D: Practice ideas for the weekend

- Build a FastAPI app with GET /quotes and GET /quotes/random returning motivational quotes.
- Add POST /quotes that returns 201, and return 404 for unknown ids.
- Call a free public API with requests and print three fields from the response.

Coming up on Day 6: routing, path and query parameters, Pydantic models, full CRUD, and an introduction to ORMs with SQLAlchemy.

Yoi shuumatsu wo (良い週末を): have a great weekend